MINOR PROJECT

DIABETES PREDICTION USING GRADIENT BOOSTING

AIML Minor Project

Submitted by

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Batch - A

1. Introduction

Diabetes is a major health-related condition that can be studied using data-driven methods. This project uses a Kaggle diabetes dataset to explore patient-related information and build a machine learning model that predicts whether a person is likely to have diabetes. The project follows an end-to-end workflow covering data understanding, cleaning, exploratory data analysis, preprocessing, model training and evaluation.

2. Problem Statement

The objective is to develop a machine learning classification model that can predict the diabetes status of a person using available demographic and health-related features. The project also aims to identify useful patterns in the data through exploratory analysis.

3. Project Objectives

• Understand the structure and quality of the Kaggle dataset.
• Clean and prepare the data for machine learning.
• Explore important variables using meaningful visualizations.
• Train a Gradient Boosting classification model.
• Evaluate the model using suitable classification metrics.
• Summarize findings, limitations and possible future improvements.

4. Dataset Description

The project uses the **Diabetes Prediction Dataset** obtained from Kaggle. The dataset contains 100,000 records and 9 variables. The target variable is **diabetes**, where the model learns to distinguish between diabetes and non-diabetes cases. Important input variables include gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level and blood glucose level.

5. Data Understanding and Cleaning

The dataset was loaded into Python using Pandas. The shape, column names, missing values and duplicate records were checked. Duplicate rows were removed where present. The categorical variables, including gender and smoking history, were converted into numerical features using one-hot encoding so that they could be used by the machine learning model.

6. Exploratory Data Analysis

Exploratory data analysis was performed to understand the distribution of the target and important numerical features. Five meaningful visualizations were created: diabetes distribution, age distribution, BMI by diabetes status, blood glucose level by diabetes status, and HbA1c level by diabetes status. These plots help identify differences and patterns between people with and without diabetes.

7. Machine Learning Approach

The project uses a **Gradient Boosting Classifier**. Gradient Boosting is an ensemble learning technique that builds a sequence of decision trees, with later trees focusing on errors made by earlier trees. It is

suitable for a binary classification problem such as predicting the diabetes target variable.

8. Model Preparation

The encoded dataset was divided into input features (X) and the target variable (y). The data was split into training and testing sets using an 80:20 ratio. Stratified splitting was used so that the class distribution of the target was preserved in both sets. The Gradient Boosting model was then trained using the training data.

9. Model Evaluation

The trained model was evaluated on the test data. Accuracy, precision, recall and F1-score were used to assess the classification performance, and a confusion matrix was generated to show correct and incorrect predictions. The final accuracy and detailed classification results are available in the project notebook and should be reported directly from the executed notebook output.

10. Feature Importance

Feature importance was examined from the trained Gradient Boosting model. This helps identify which input variables contributed most to the model's predictions. The feature-importance visualization provides an interpretable summary of the relative contribution of the available features.

11. Conclusion

This project used the Kaggle Diabetes Prediction Dataset to study and predict diabetes using machine learning. The data was cleaned and analyzed to understand important patterns and relationships between the features and diabetes. Visualizations helped explore factors such as age, BMI, HbA1c level and blood glucose level. A Gradient Boosting Classifier was then trained to predict diabetes status and evaluated using standard classification measures and a confusion matrix. The project demonstrates how data analysis and boosting techniques can be applied to a healthcare-related prediction problem.

12. Limitations

The model is intended as an academic machine learning project and should not be treated as a medical diagnostic tool. The dataset represents the information available in the source dataset and may not capture every factor involved in real clinical diagnosis. Model performance can also vary when applied to different populations or new data.

13. Future Improvements

Future work could compare Gradient Boosting with other boosting algorithms, perform systematic hyperparameter tuning, use additional evaluation techniques, and test the model on independent real-world datasets. The project could also be extended into a user-friendly prediction application for educational purposes.

14. Kaggle Dataset

Dataset source: Kaggle – Diabetes Prediction Dataset.
The dataset was selected according to the project requirement to use a suitable Kaggle dataset for an end-to-end machine learning project.

15. Final Summary

The project completed the main stages of an end-to-end machine learning workflow: defining the problem, understanding and cleaning the data, performing exploratory analysis, preparing features, training a Gradient Boosting model, evaluating the predictions, and interpreting the results. The notebook contains the executable implementation and visualizations, while this report provides a concise explanation of the project.